NAME : SUDEEP J

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CAREER OBJECTIVE

Embedded Software Engineer fresher with practical experience in Embedded C and bare-metal microcontroller programming. Worked on peripheral interfacing, interrupt handling, and debugging firmware using industry-standard tools. Looking for an entry-level role in a core product company to work on real embedded firmware and hardware–software integration.

WORK EXPERIENCE

- Completed technical training internship – Advanced Embedded Systems Course at Emertxe Information Technologies (http://www.emertxe.com), Bangalore

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TECHNICAL SKILLS

- Programming Languages:
    - **Embedded C (Basic)**
    - **Assembly Language (8051 and arm cortex M3)**
    - **Verilog (RTL design – basics)**
- Embedded controllers:
    - Hands-on working with GPIOs, Analog I/Os, Memory usage, interfacing, character LCD
    - Peripherals usage - Timers, Counters and Interrupts
    - Communication protocols - UART, SPI, I2C etc
- Embedded platforms and tools:
    - Linux (Ubuntu)
    - Proteus 8 Professional
    - Wowki online simulator
    - MATLAB V2023 and V2025
    - Keil version 3 and 4 (ARM Cortex M3)
    - MPLAB and PICSIMLAB
- Microcontroller Boards:
    - 8051 and arm cortex M3
    - Ardiuno uno R3
    - ESP32

COURSE WORK

- Operating Systems (Basics)
- Digital Electronics
- Digital Signal Processing

EDUCATION	• B.T (ECE), GM University, Davangere, 2024-2027 • Diploma – (ECE), Government polytechnic Harihar ,94.55 %, 2023 • Class – X, GM Halamma high school,Davangere 66.55%,2018
CONTRIBUTIONS	• Completed an **Embedded Systems Internship** with hands-on experience in microcontroller programming, hardware interfacing, and debugging embedded applications. • Participated in **Smart India Hackathon (SIH)**, gaining experience in rapid problem-solving, team-based development, and real-world engineering challenges. • Served as an **NSS Cadet**, contributing to social outreach programs and demonstrating teamwork and organizational discipline.

PROJECTS AT EMERTXE

Project Number:1

Title	**Microcontroller-Based Microwave Oven Controller**
Project brief	Designed and implemented an embedded control system for a **microwave oven** using a microcontroller.The system handles **user input through a keypad**, displays real-time status on an **LCD**, manages **timing operations**, and controls different operating modes using a **state-machine-based firmware architecture**.
Technologies used	• Embedded C – • PIC Microcontroller • MPLAB X IDE • XC8 Compiler • LCD Interface • Matrix Keypad • Timers & Interrupts
Key challenges & Learnings	✓ Designing a clear **state machine** for multiple oven modes ✓ Achieving **accurate timing** using hardware timers and ISRs ✓ Formatting **CLCD display output** within limited memory ✓ Debugging firmware logic during simulation
GitHub Link Repository:	https://github.com/sudeep-embedded/embedded-systems-portfolio/tree/main/projects/proteus-projects/Emertxe-Internship